

An Introduction to 3D Printing

Part 2: Practical Application

Examples, Critical Thinking, and Design

Outline

Part 1: Printers

Process

Limitations

Dimensional Accuracy

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

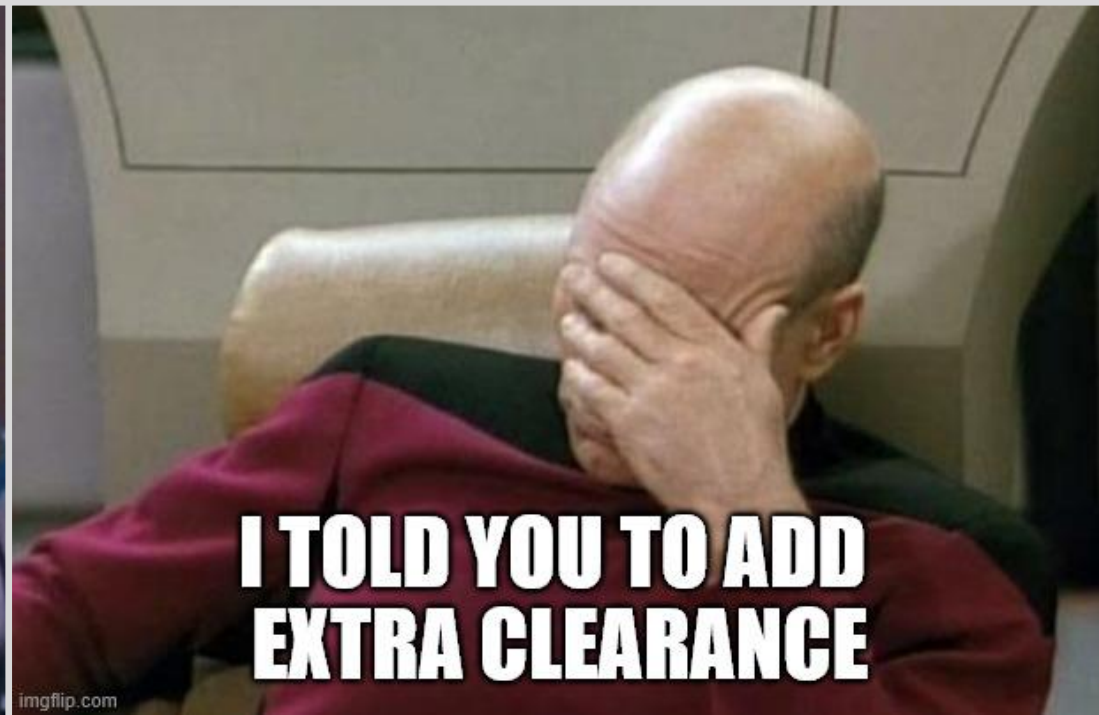
Vibrations and Waves

Linear Motion

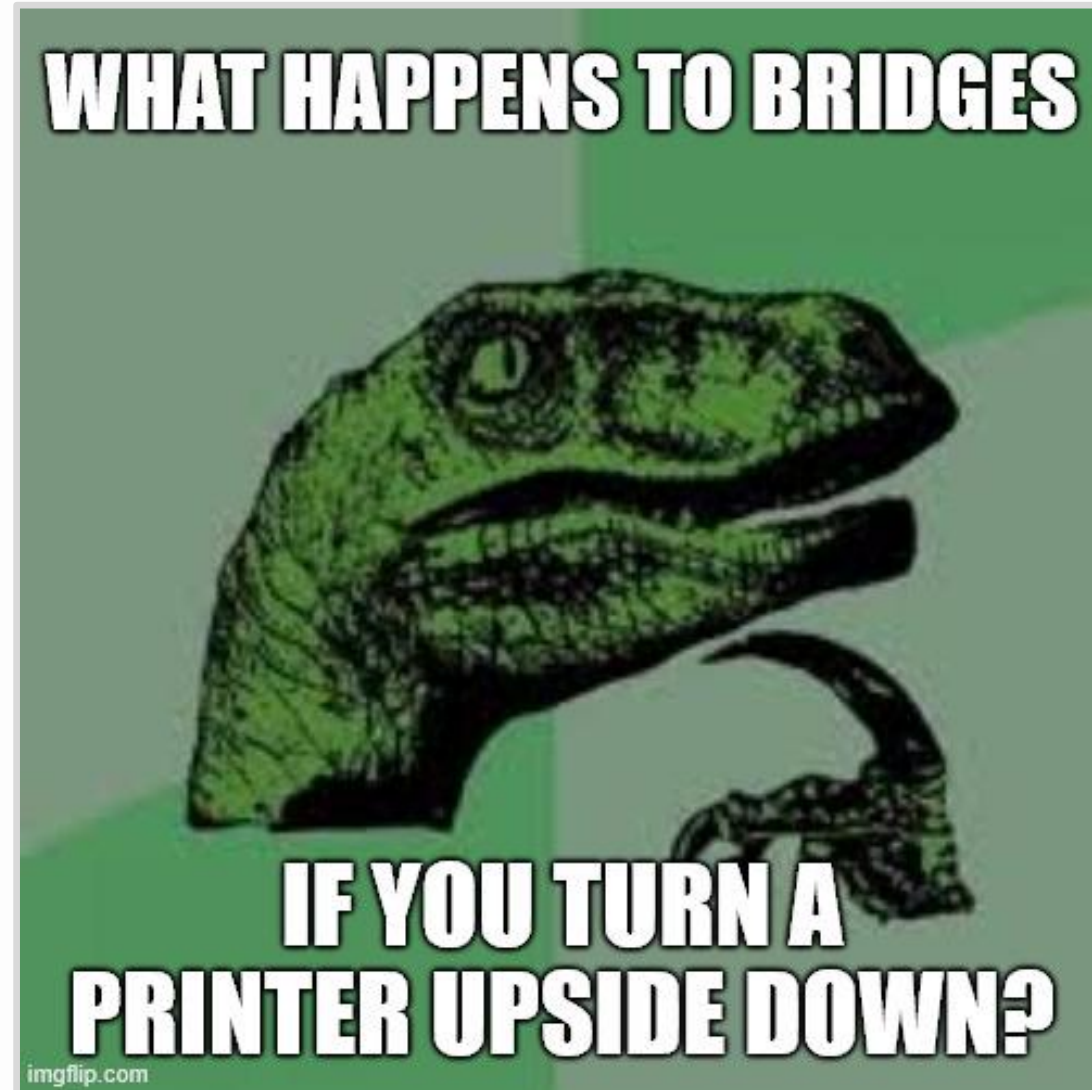
Thermal Expansion

Pop Quiz

- How can you help printed parts fit together?
 - You sometimes need scale factor.
 - You always need constant offset.



Anti-Gravity



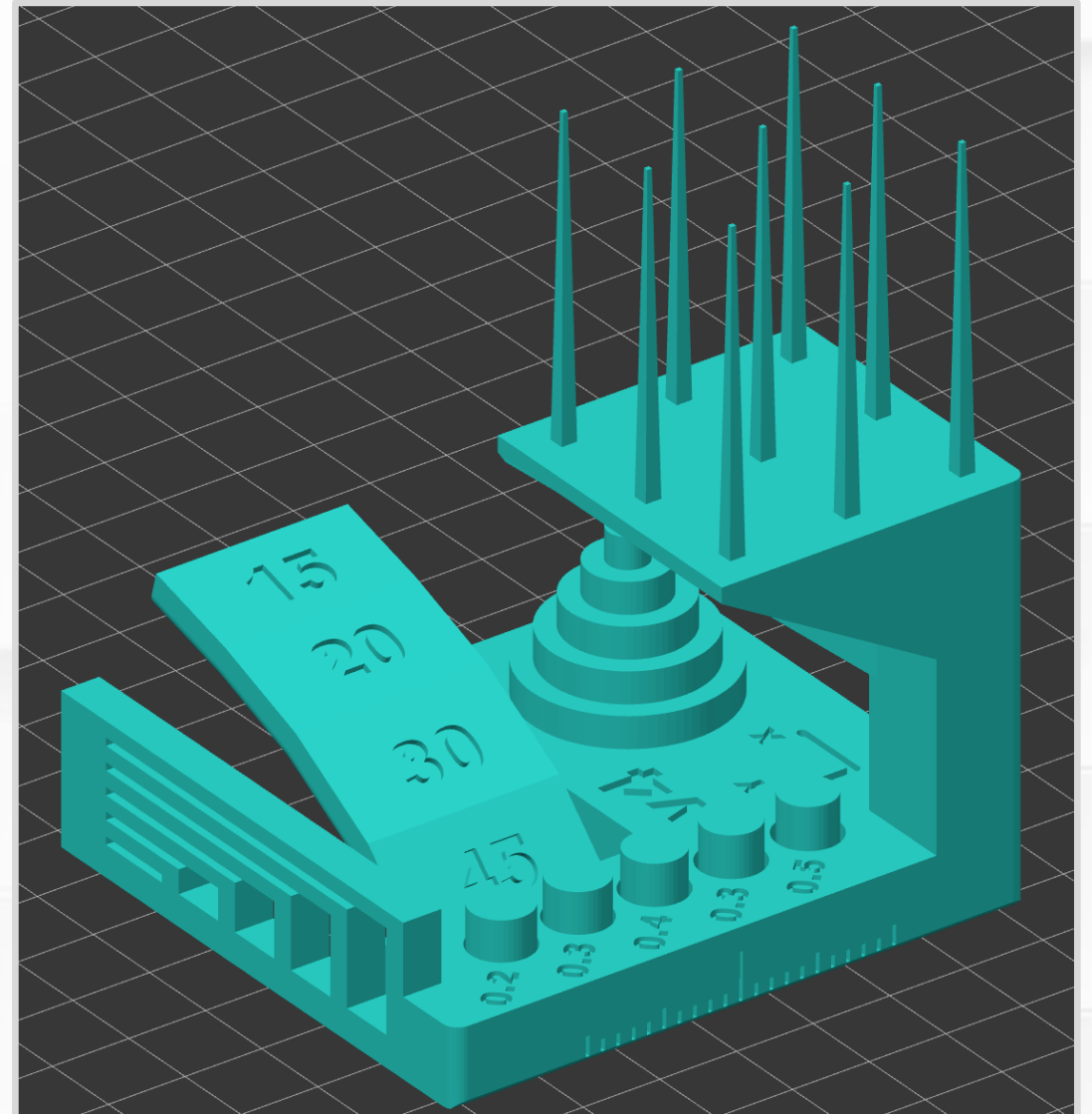
Anti-Gravity

- Positron Github.
 - Prints upside down.
 - Folds up and fits in a small box.
- Design team says ... same print quality.
- Why?
- Extrusions hold their shape when you press them onto a solid surface.



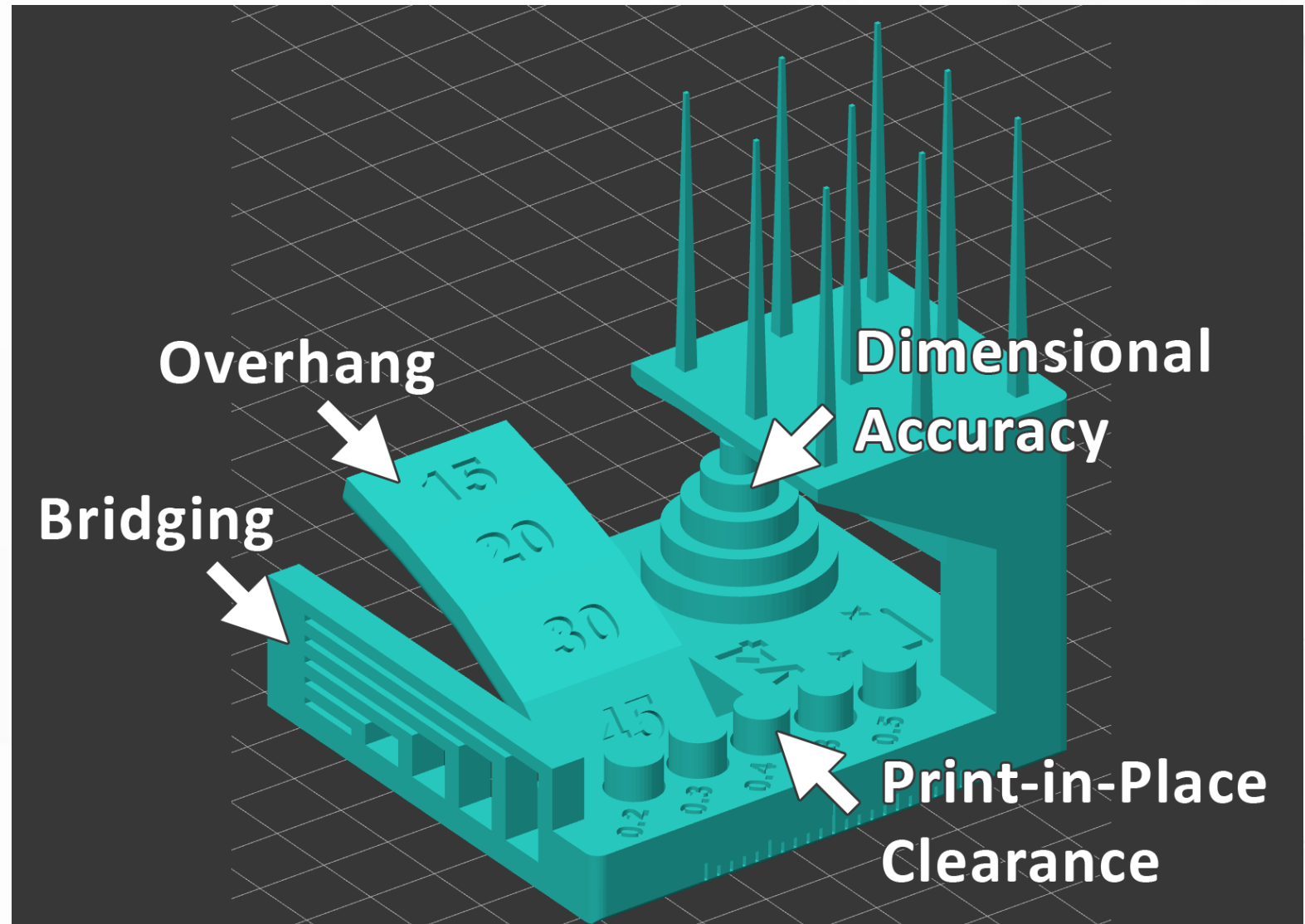
Critical Thinking: Stress Testing

- Autodesk Test Print. One of many test prints.
- Prints that highlight the **capabilities** and **limitations** of your printer + filament.
- What are these shapes testing?
- Full breakdown [here](#).



Critical Thinking: Stress Testing

- Here are limitations covered in this talk.



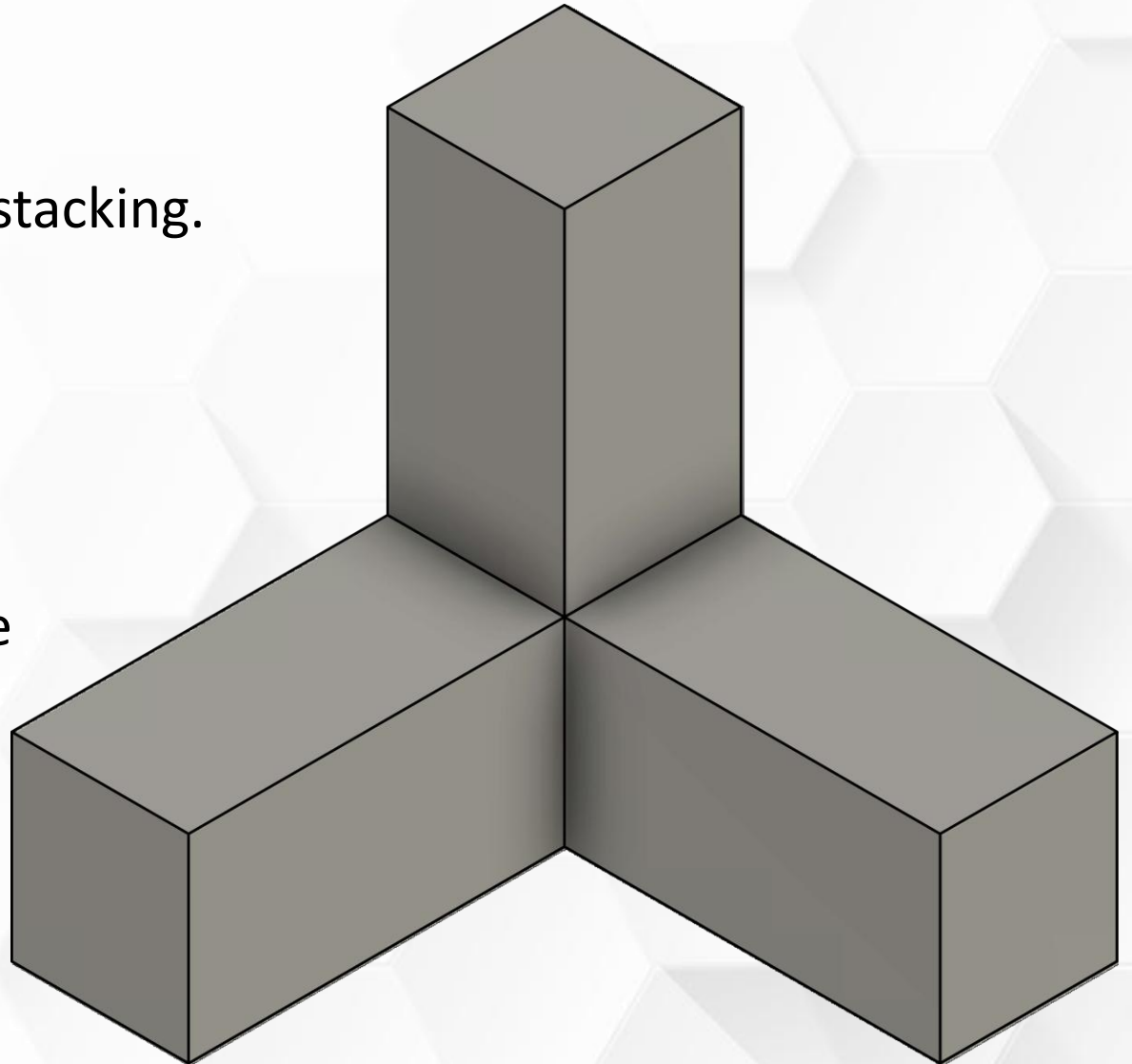
Critical Thinking: Corners

- Rounding corners with fillets is a good idea. Why?
- Stresses concentrate in corners.
 - Adding fillets improves strength.
 - Fillets reduce first layer warping.
- Print faster with more accurate movements. Why?
 - Think of a racecar (toolhead) going around a track.
 - Smoothing changes of direction reduces vibrations.

Critical Thinking: Corner Bracket

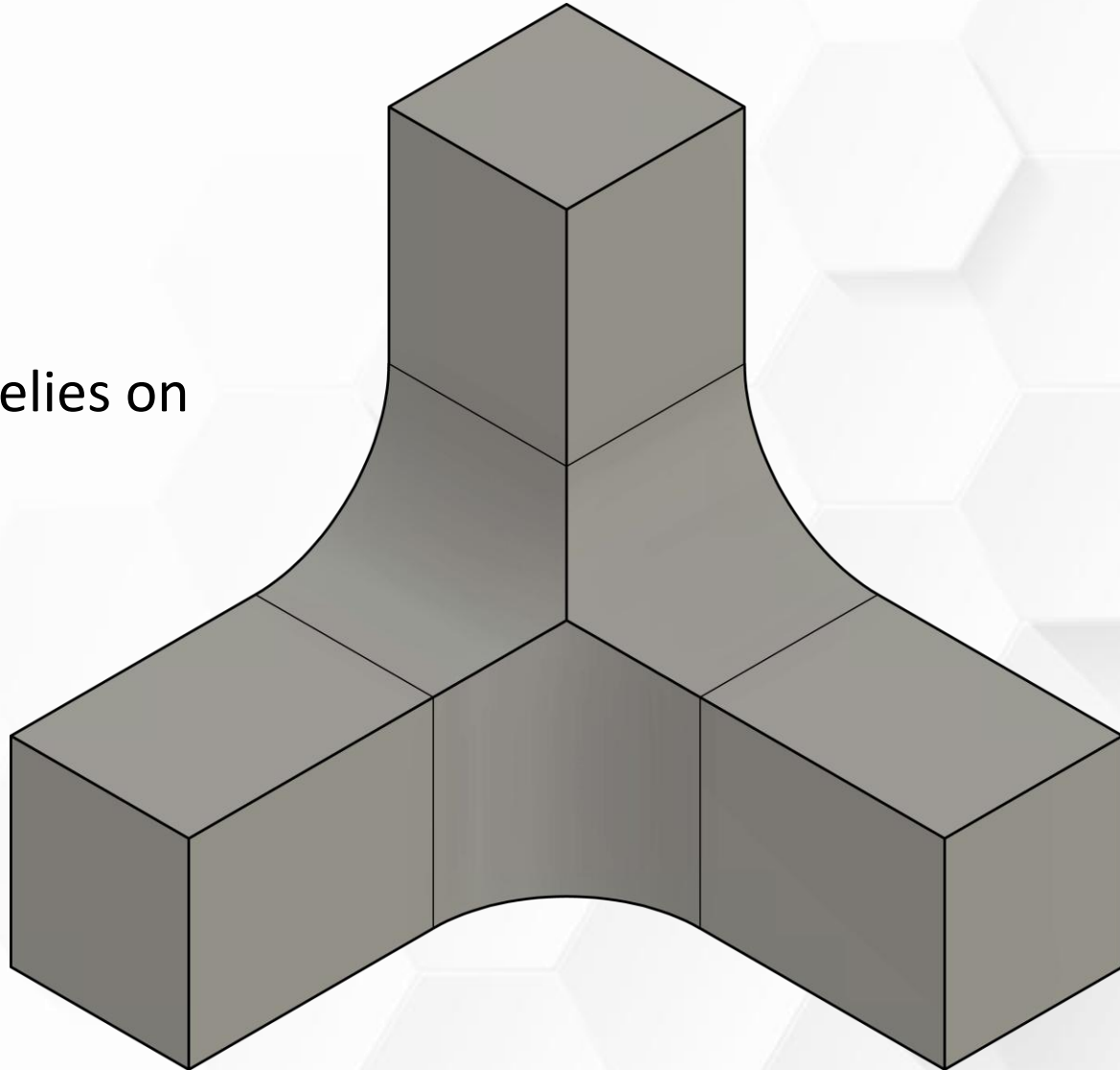
Consider this corner bracket.

- Top section is weaker because of layer stacking.
- How can you improve strength?
 - Stronger filament.
 - Stronger slicer settings.
 - What else?
- Let's rethink the design and make some modifications.



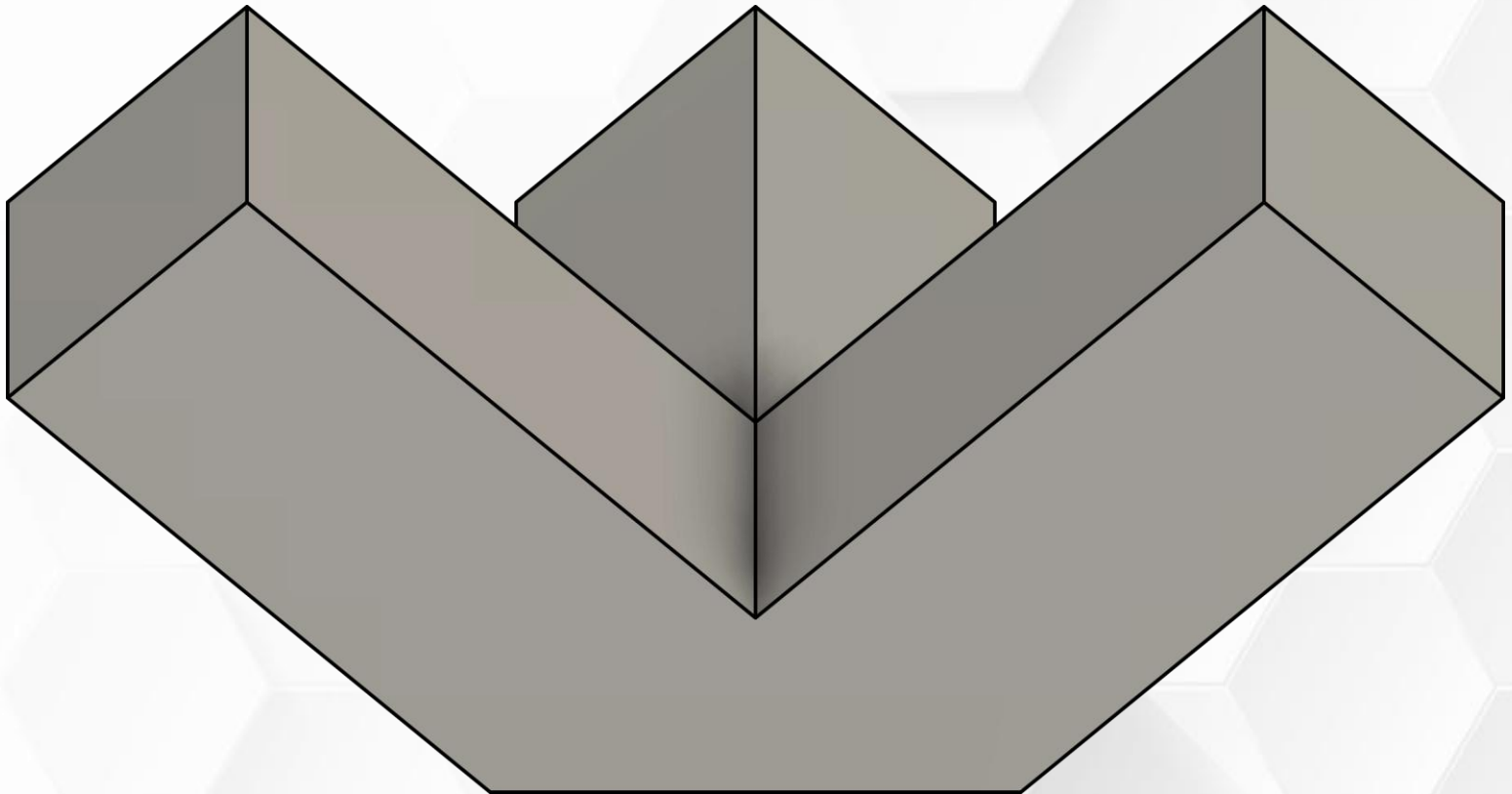
Critical Thinking: Corner Bracket

- Reinforce joints with **fillets**.
- Reduces stress concentration.
- Removes the weakest point.
- Improved strength, but top still relies on vertical layer stacking.



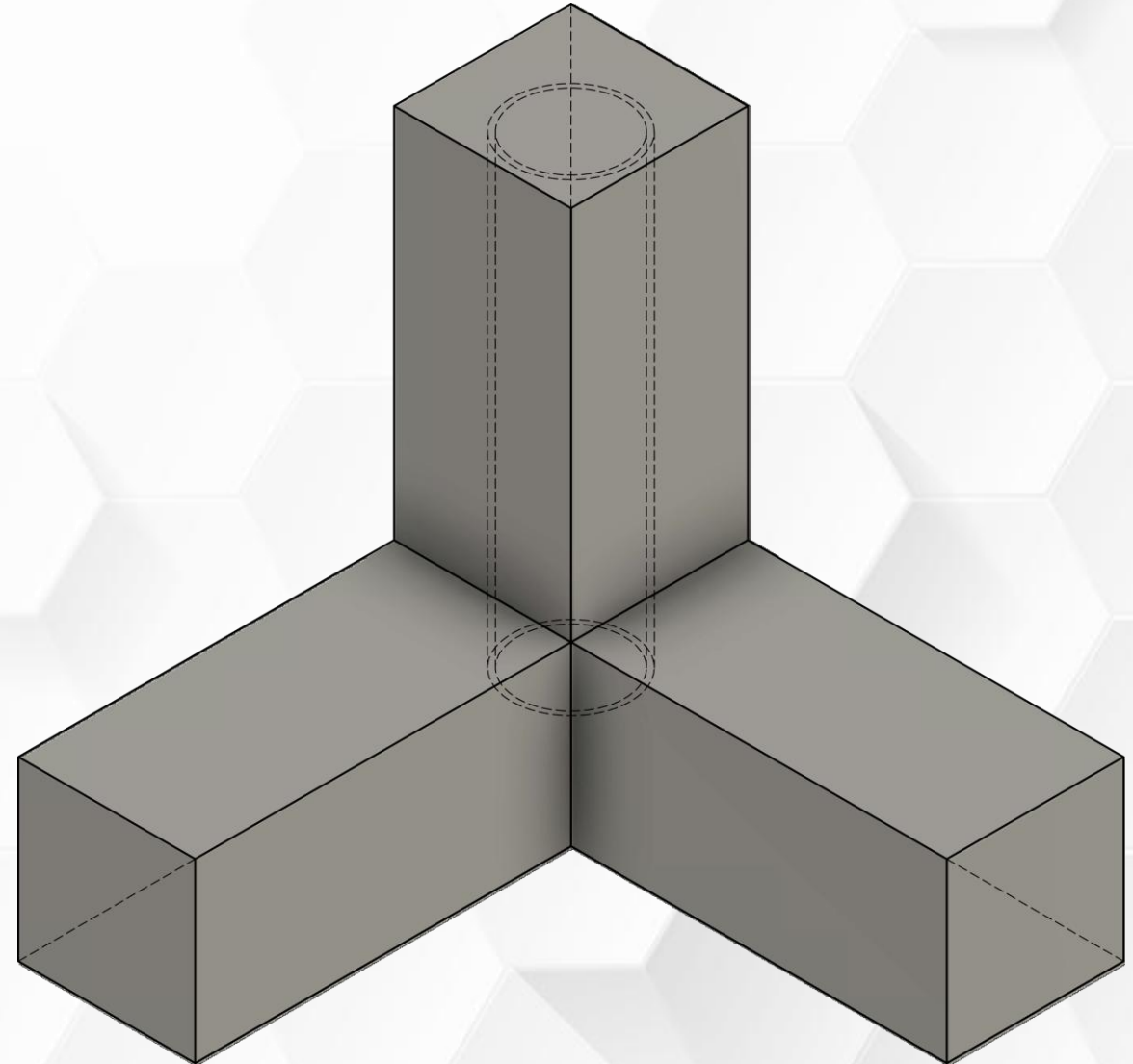
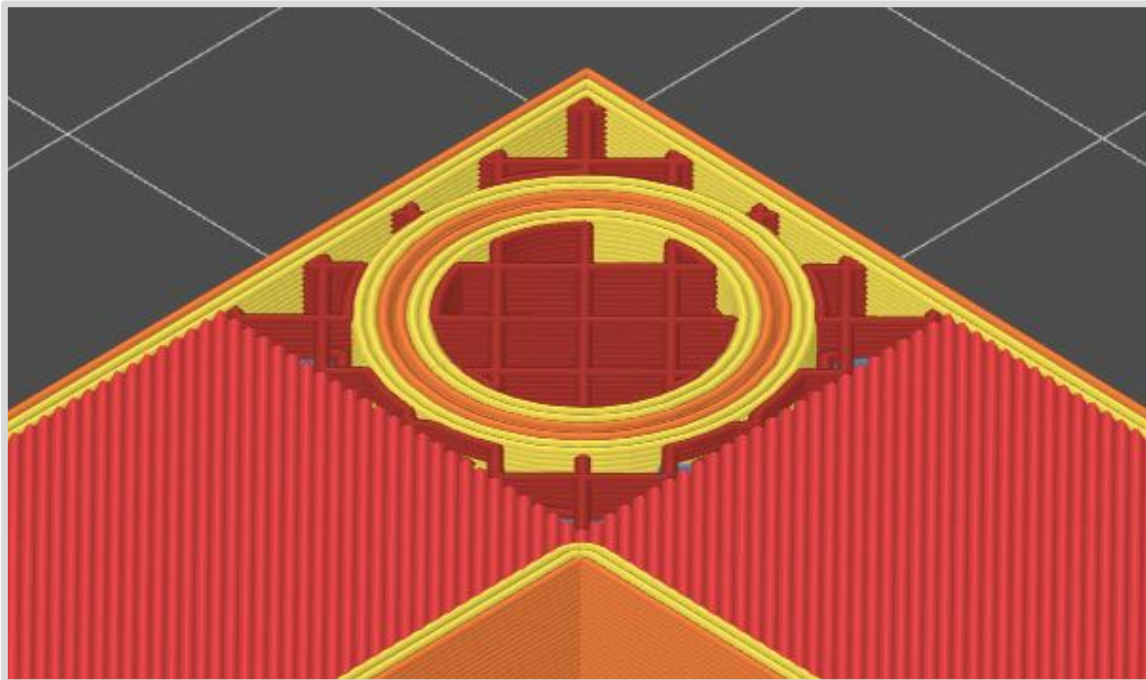
Critical Thinking: Corner Bracket

- Trim the corner to create an angled print orientation.
- Adds overhangs.
- Reduced first layer contact.
- Each section has equal strength.
- Improved strength, depending on direction of forces.



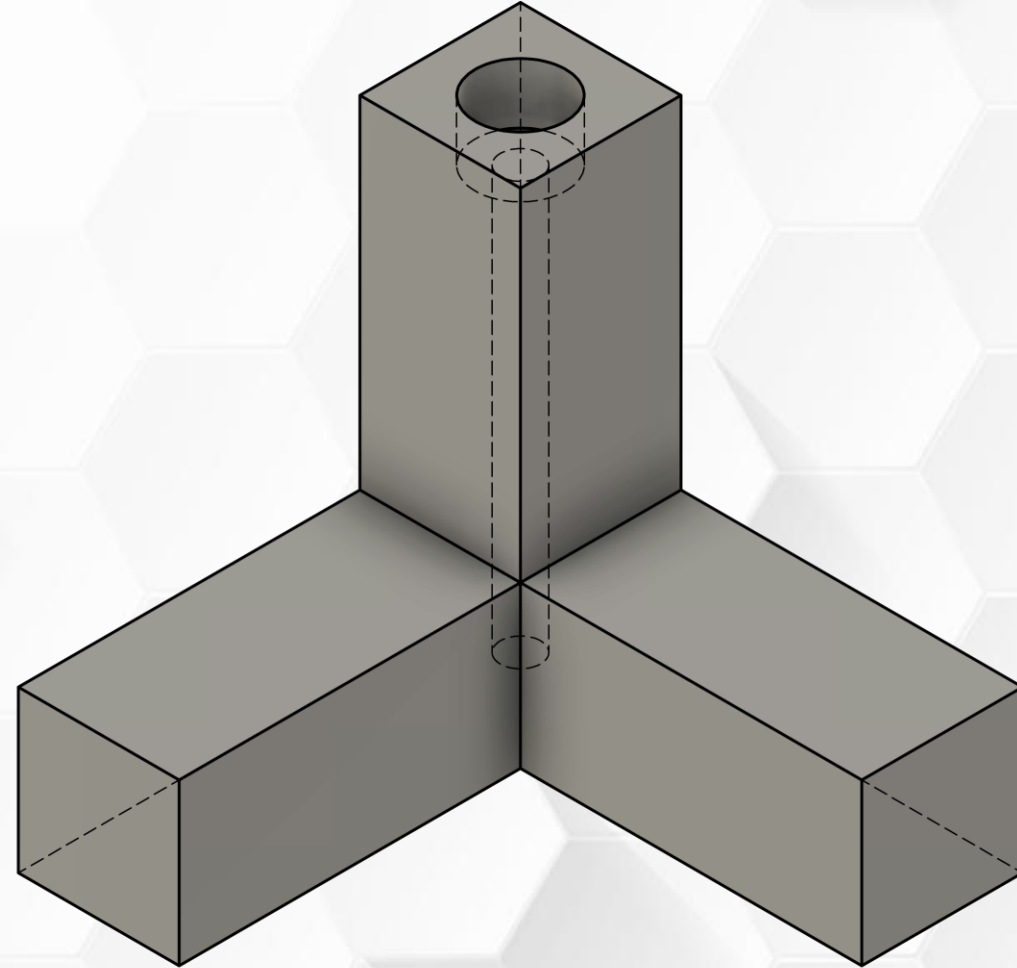
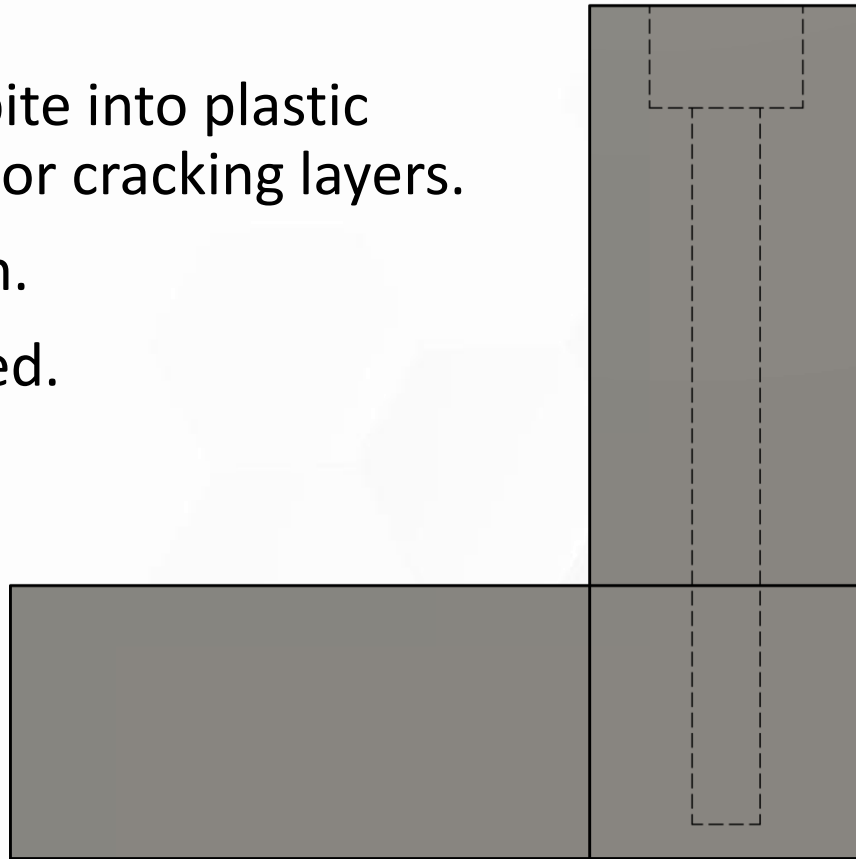
Critical Thinking: Corner Bracket

- Cut a small slit (around 0.1 mm wide).
- Tricks slicer into adding extra perimeters.
- Generally, perimeters are stronger than infill.
- Improved strength and looks the same.



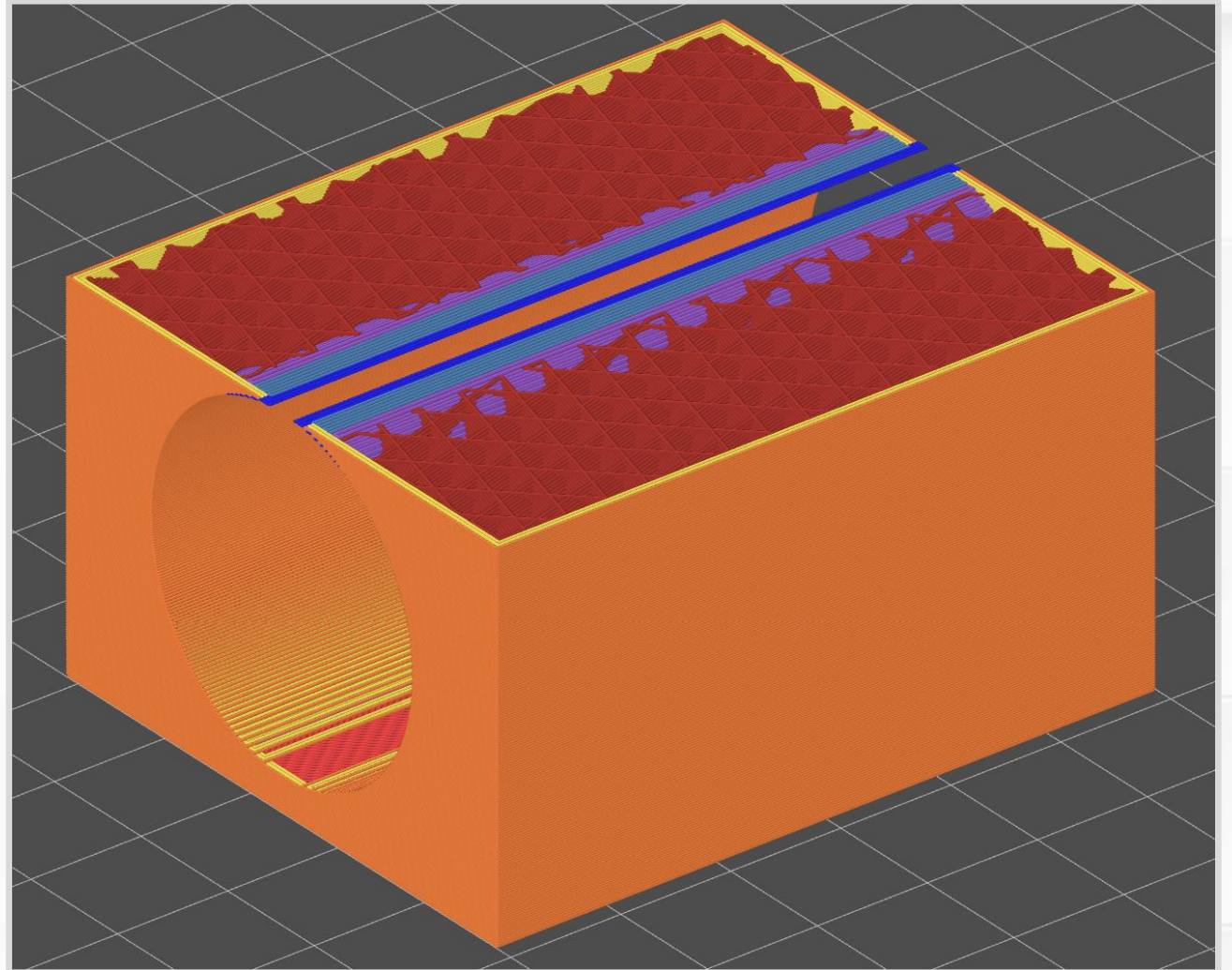
Critical Thinking: Corner Bracket

- Cut a hole. Add tight fitting screw after printing.
- Threads should bite into plastic without splitting or cracking layers.
- Superior strength.
- Assembly required.



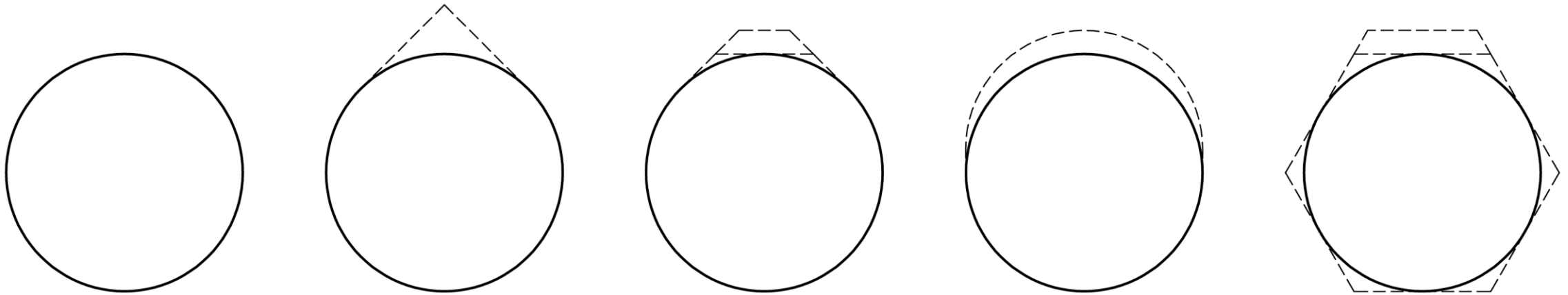
Critical Thinking: Horizontal Holes

- Horizontal holes have extra steep overhangs at the top.
- If extrusions collapse, the hole is undersized.
- Assume the hole must remain horizontal.
- How can you alter the design to fix the problem?



Critical Thinking: Horizontal Holes

- Make the top profile more printer-friendly.
- Add a clearance at the top.



Critical Thinking: Bridges

- What can you do to improve bridges, overhangs, and floating extrusions?
(Besides supports.)
- Help extrusions cool and lock in place.
 - Lower filament temperature.
 - Slower printing speed.
 - Increase cooling.
- What if there isn't a good print orientation?
 - Reimagine the object as multiple pieces.
 - Each piece is free from supports and bridges.

Sequential Bridges

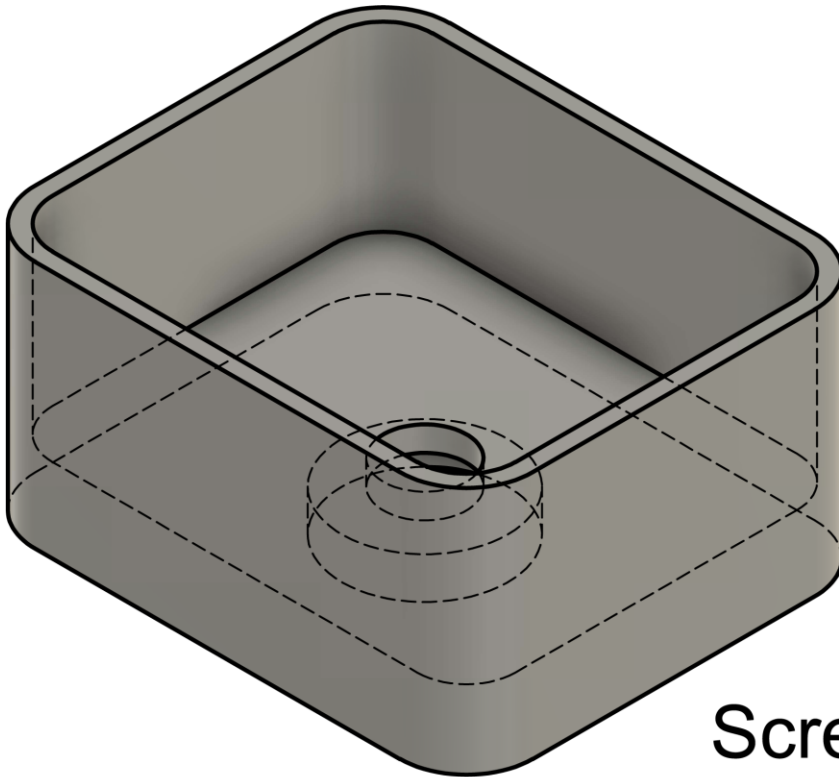
Let's combine small bridges to create a large, multi-step bridge.



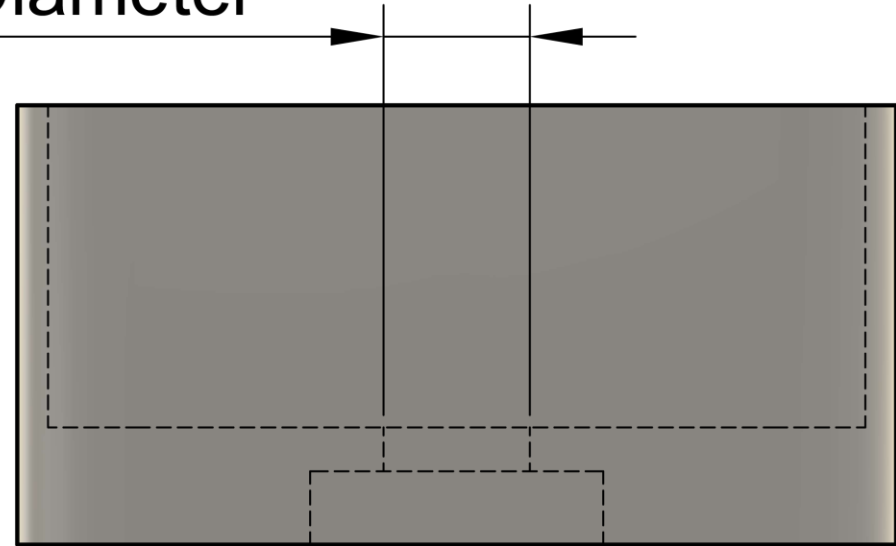
Critical Thinking: Sequential Bridges

Consider a storage box with a countersunk screw on the bottom.

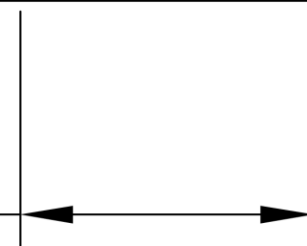
- Can you see the hidden problem?



Screw Diameter



Screw Head Diameter



WARNING:

Detected print stability issues:

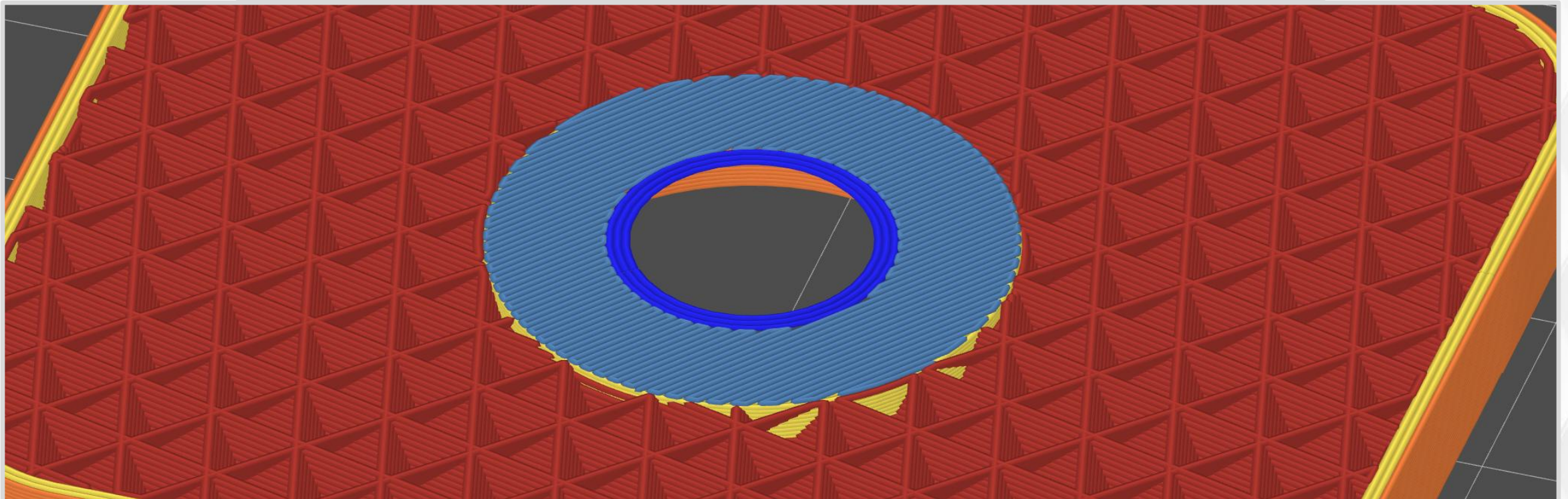


Box Bridge1

Floating bridge anchors, Long bridging extrusions

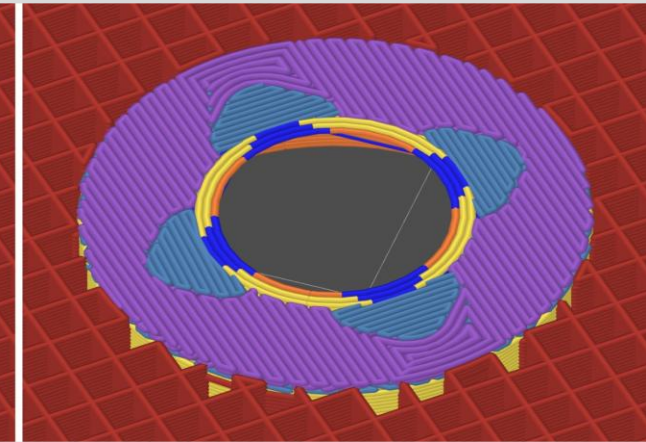
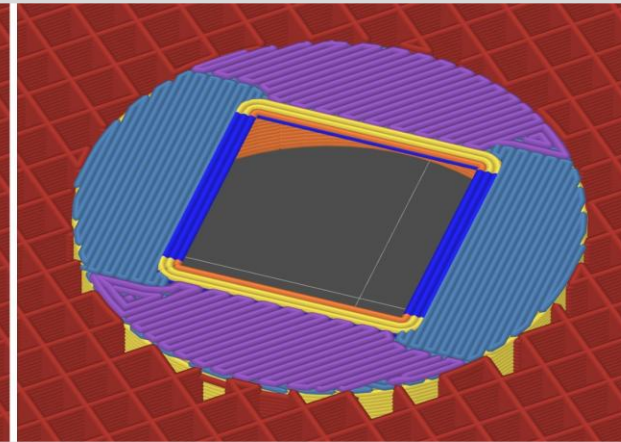
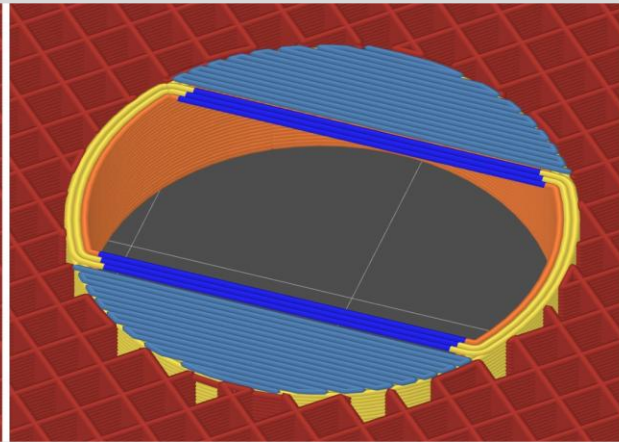
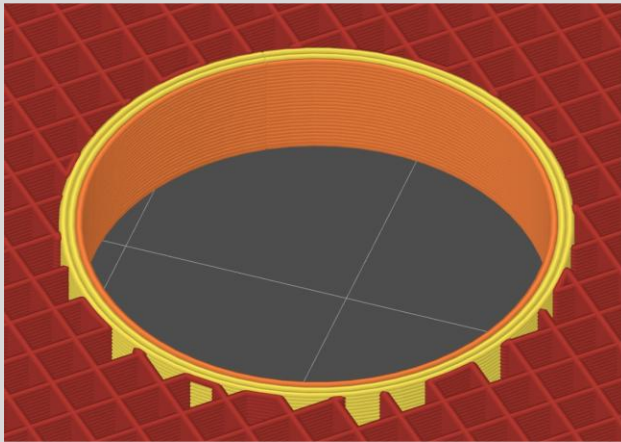
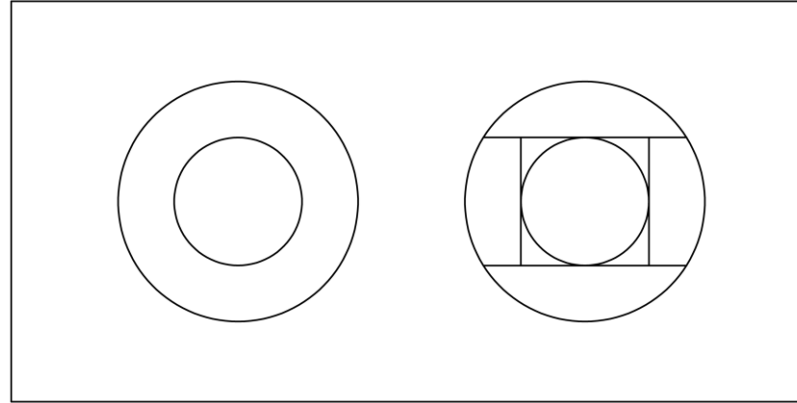


Consider enabling supports.



Critical Thinking: Sequential Bridges

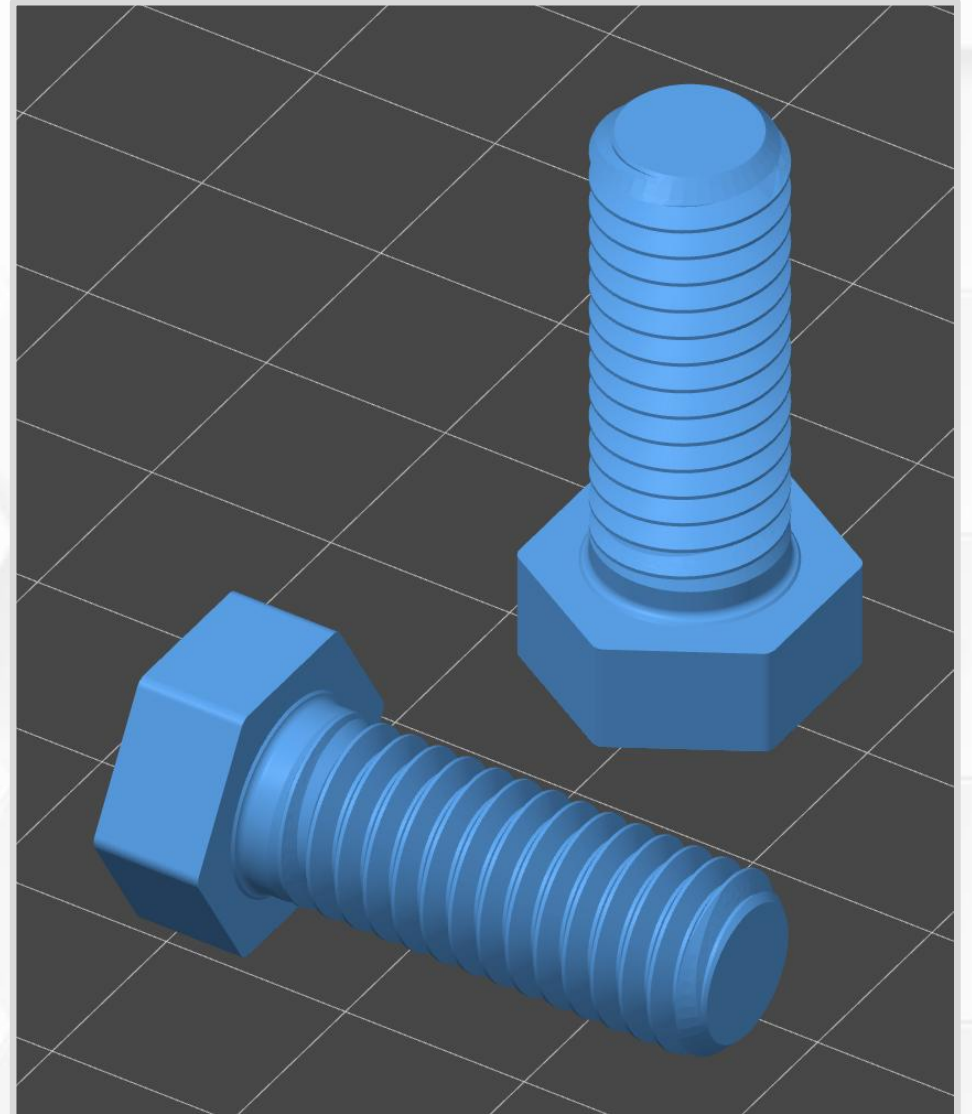
- There are floating extrusions!
- We can fix this without supports.
- Complete the bridge in multiple steps.



Critical Thinking: Screws

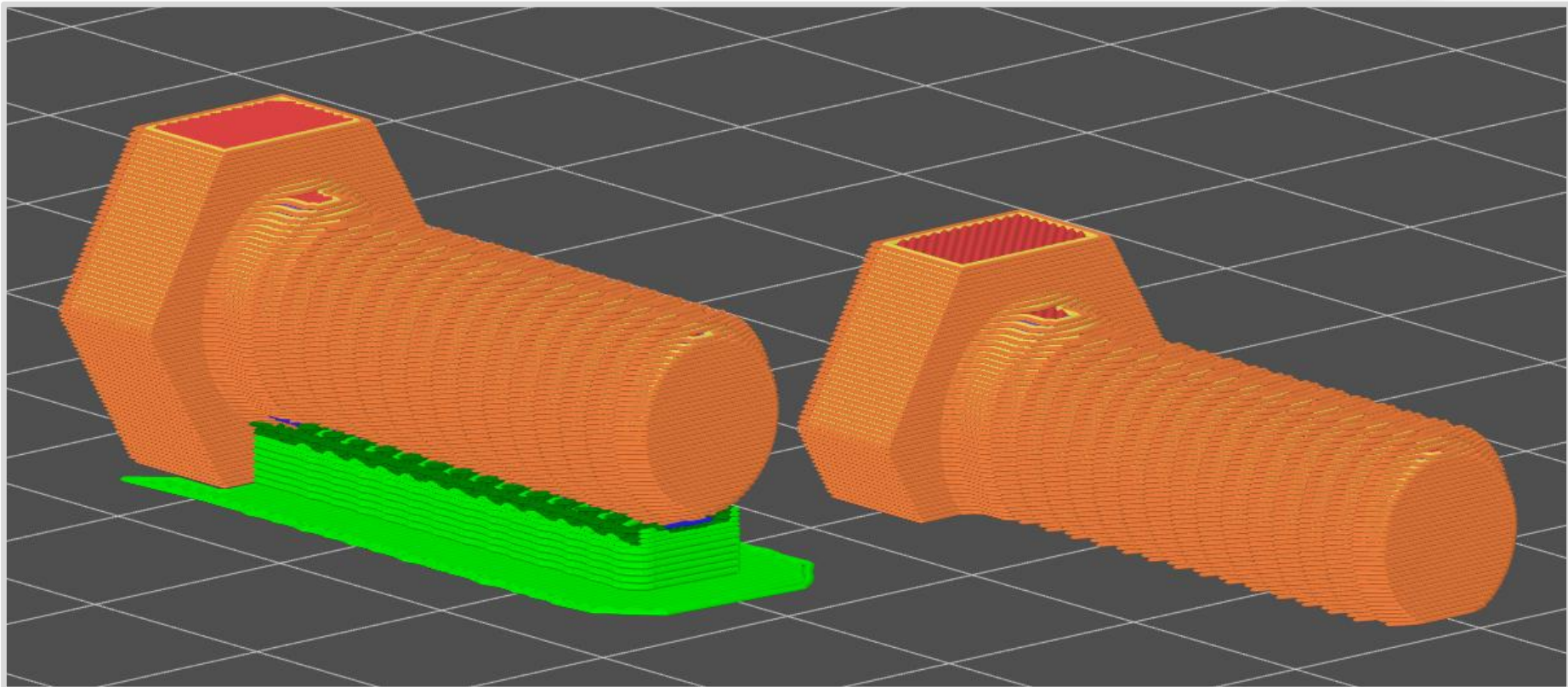
Let's 3D print a plastic screw.

- Compare and contrast print orientations.
 - What differences can you see in strength and printability?
 - Horizontal.
 - Layer lines maximize strength.
 - Nozzle must be smaller than threads.
 - Bottom threads are floating extrusions.
- Suggestions?



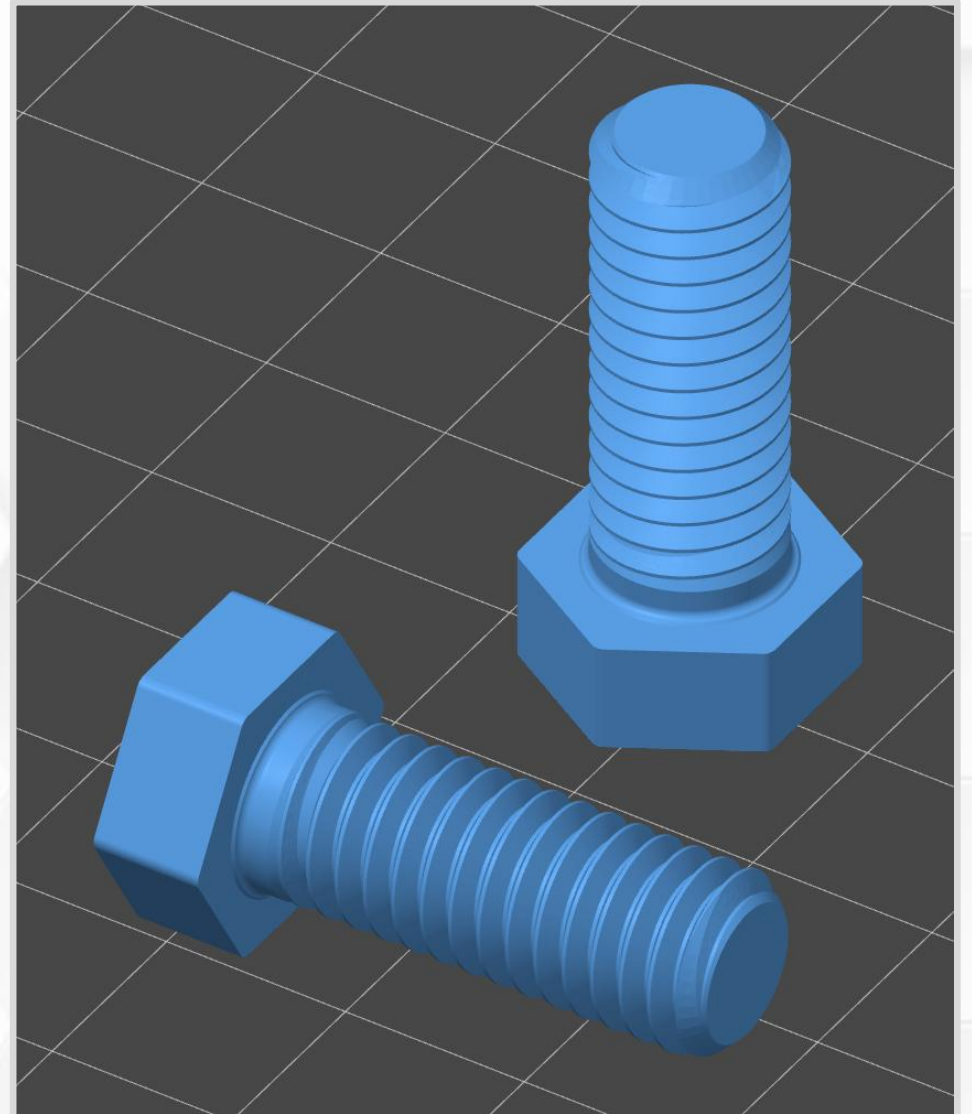
Critical Thinking: Screws

- Supports will likely leave a messy surface.
- Trim the bottom of the bolt so it can sit flat.



Critical Thinking: Screws

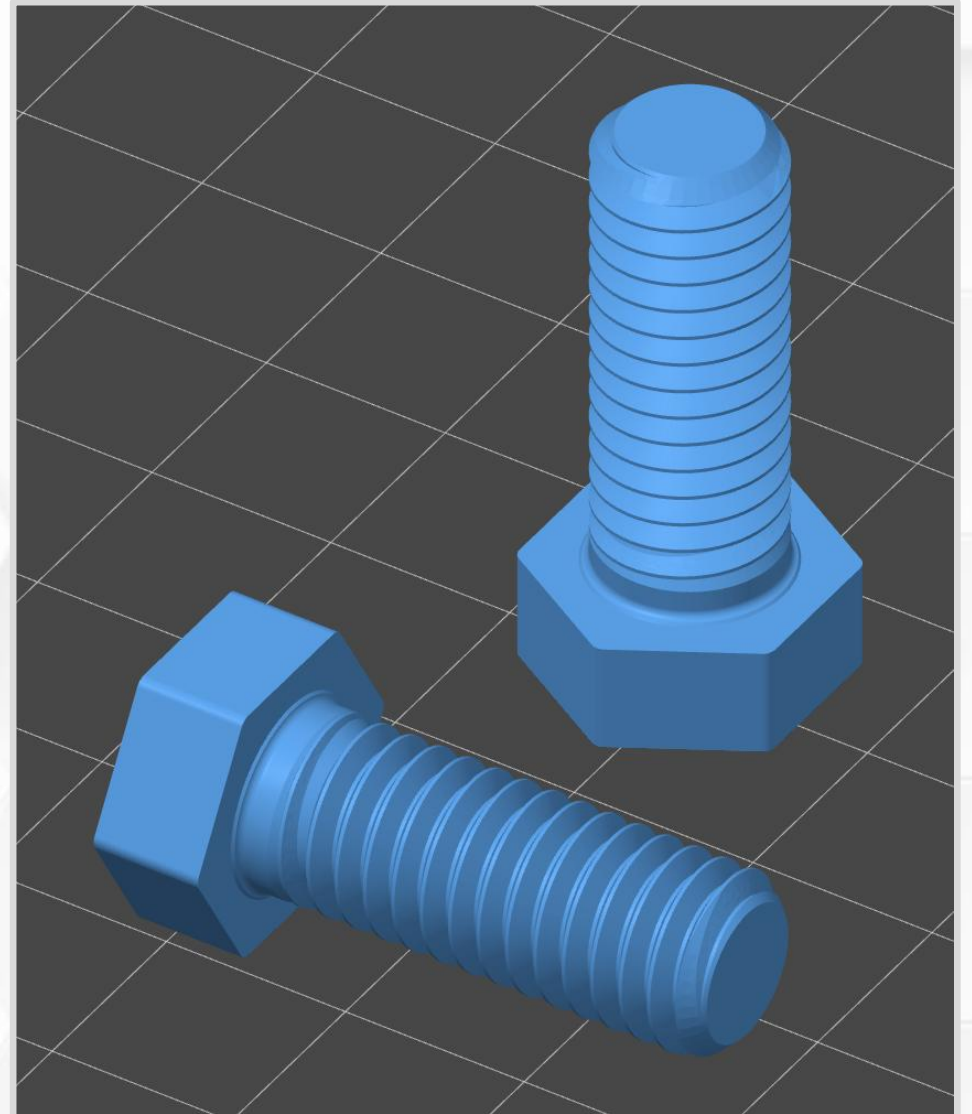
- Vertical
 - Threads limited by layer height.
 - Steep overhangs.
 - Layers lines create a weak part.
 - No supports.
 - No modifications required.



Critical Thinking: Screws

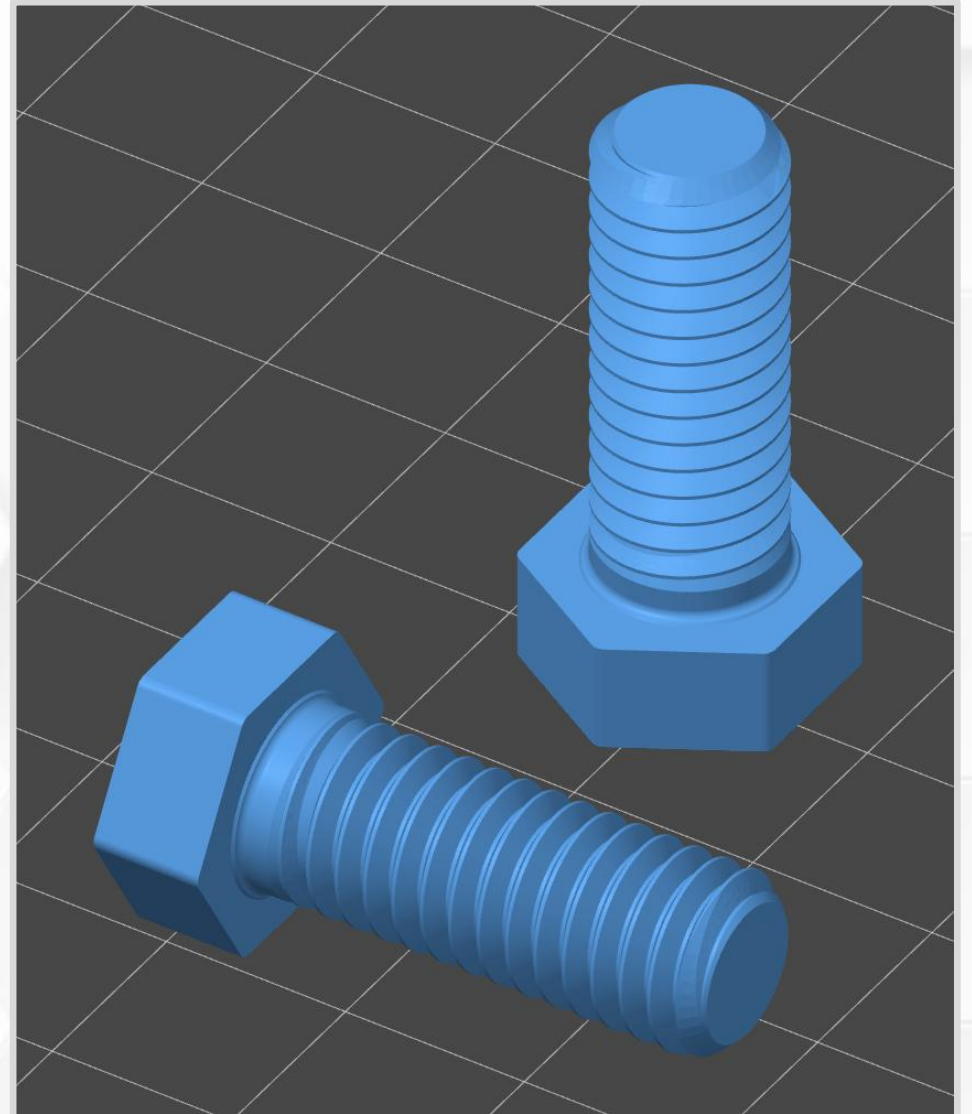
Let's summarize.

- Vertical version.
 - Easy to print.
 - Looks the best.
 - Weakest.
- Horizontal version.
 - Strongest.
 - Needs a minor modification.
 - Asymmetrical, looks weird after trimming.



Critical Thinking: Screws

- These kinds of decisions and compromises are common in 3D printing.
- Custom Threads for 3D printing by printables.com/@FlyingGyroscope.
 - Includes AutoCAD Fusion 360 lesson.
- Recommended viewing.
 - Strength testing printed bolts by My Tech Fun.
 - And a follow-up video.



Critical Thinking: Snap Buckle

Let's design a replacement part.

- Should you print a direct copy? Or does it need modifications?
- What 3D printing advantages and disadvantages do you see?
- *Hint:* Compare and contrast vertical and horizontal print orientation.



(Picture from [HomeDepot](#))

Critical Thinking: Snap Buckle

- Why is the printed buckle designed like that?



(Picture from HomeDepot)



(Snap buckle by Squinn)

Critical Thinking: Snap Buckle

- Print parts laying down.
 - High strength.
 - Perfect for the compliant mechanism.
 - Good first layer contact.
- The female socket has been reimagined.
 - The middle prong is a dovetail (trapezoid).
 - No bridges, only overhangs.

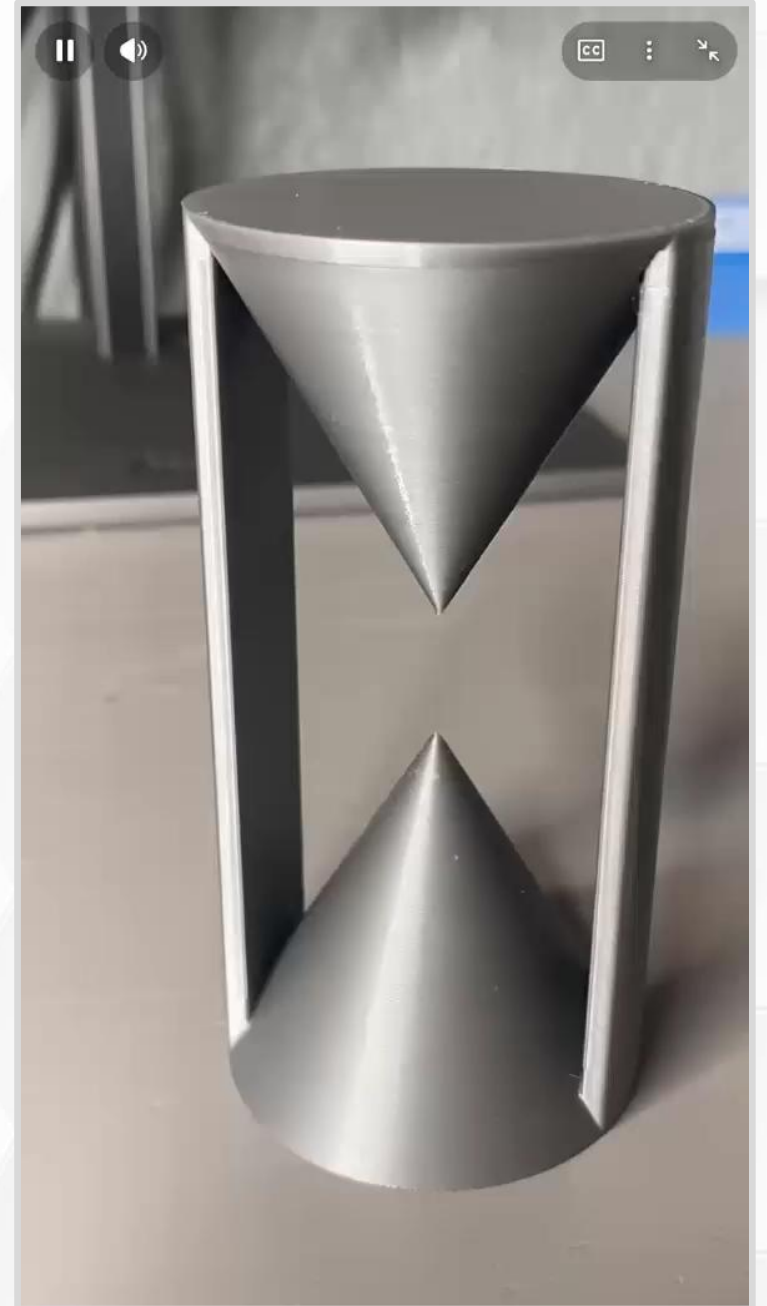


(Snap buckle by Squinn)

Creative Solutions

- I love the creativity of the maker community.
- This was printed without supports. How?
 - Put a printed part inside a printed part.

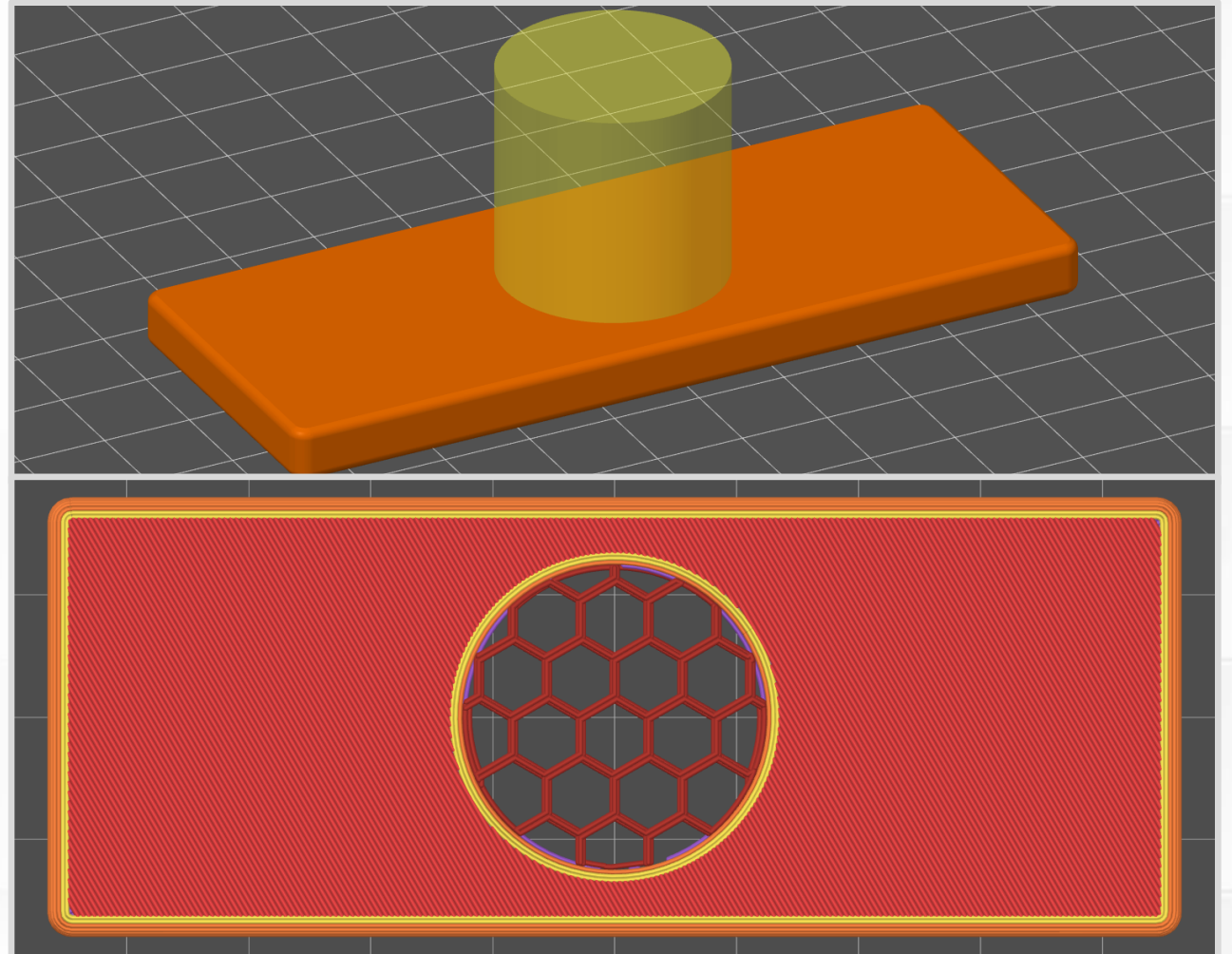
(Animation, YouTube short by [christopherhelmke](#))



Advanced Slicing: Modifiers

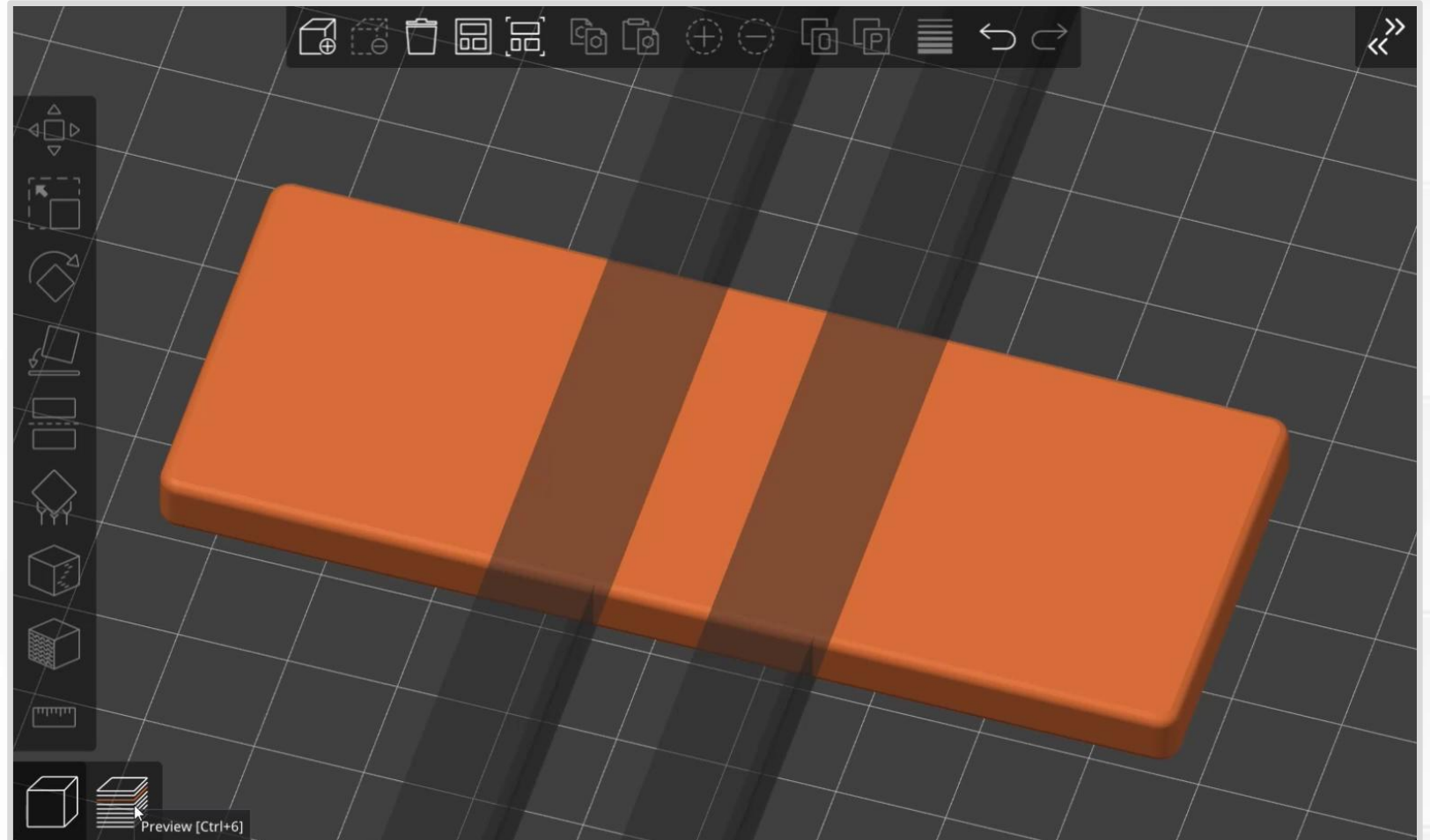
Let's make a mesh grill with PrusaSlicer.

- Add a cylinder part.
- Change it to a modifier.
- Assign zero top and bottom layers.
- Set an infill pattern.
 - Hexagon makes a nice mesh.
- Change infill percent to alter the mesh size.



Advanced Slicing: Negative Volume

- Flexify template.
Design print-in-place hinges with a slicer.
- Templates define negative space.
- Start with a flat object.
- Cut away material with templates.



(Animation, FlyingGyroscope GitHub)

